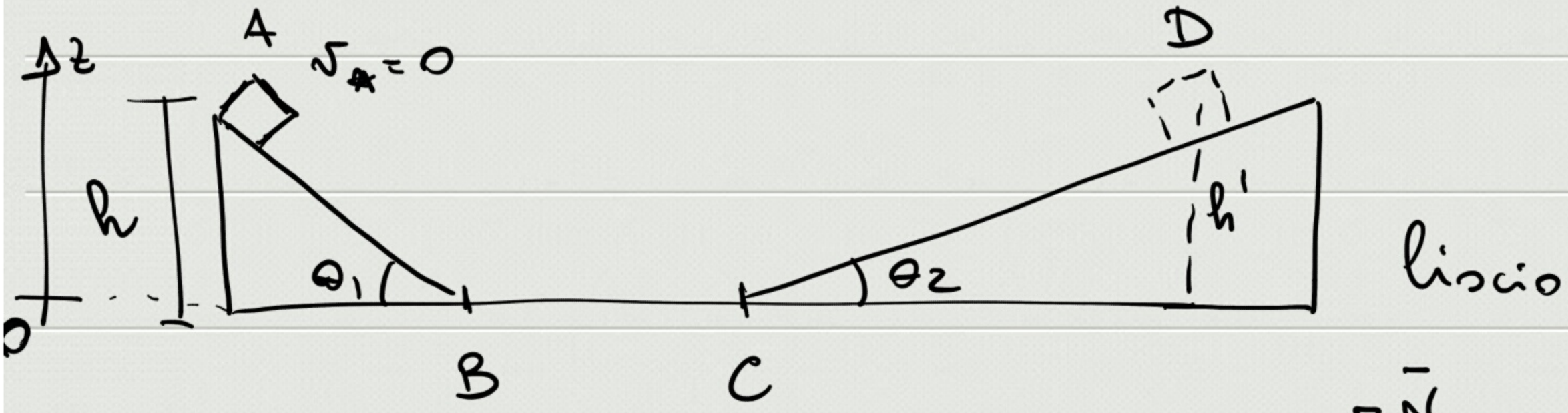
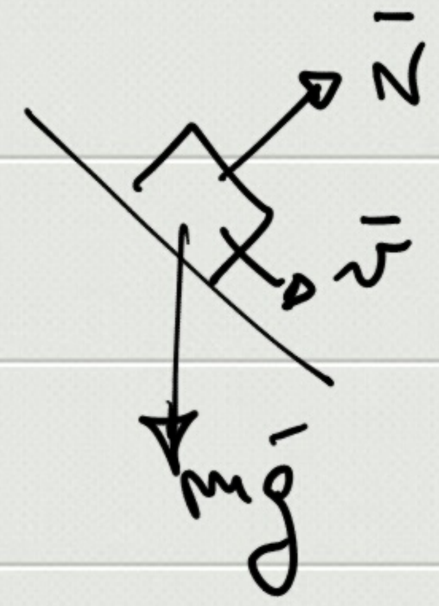


v_B, v_C, h'



forze cons. $\Rightarrow \boxed{E_m = \text{cost}}$

$$E_{m,A} = E_{k,A} + E_{p,A} = \emptyset + mgh = mgh$$

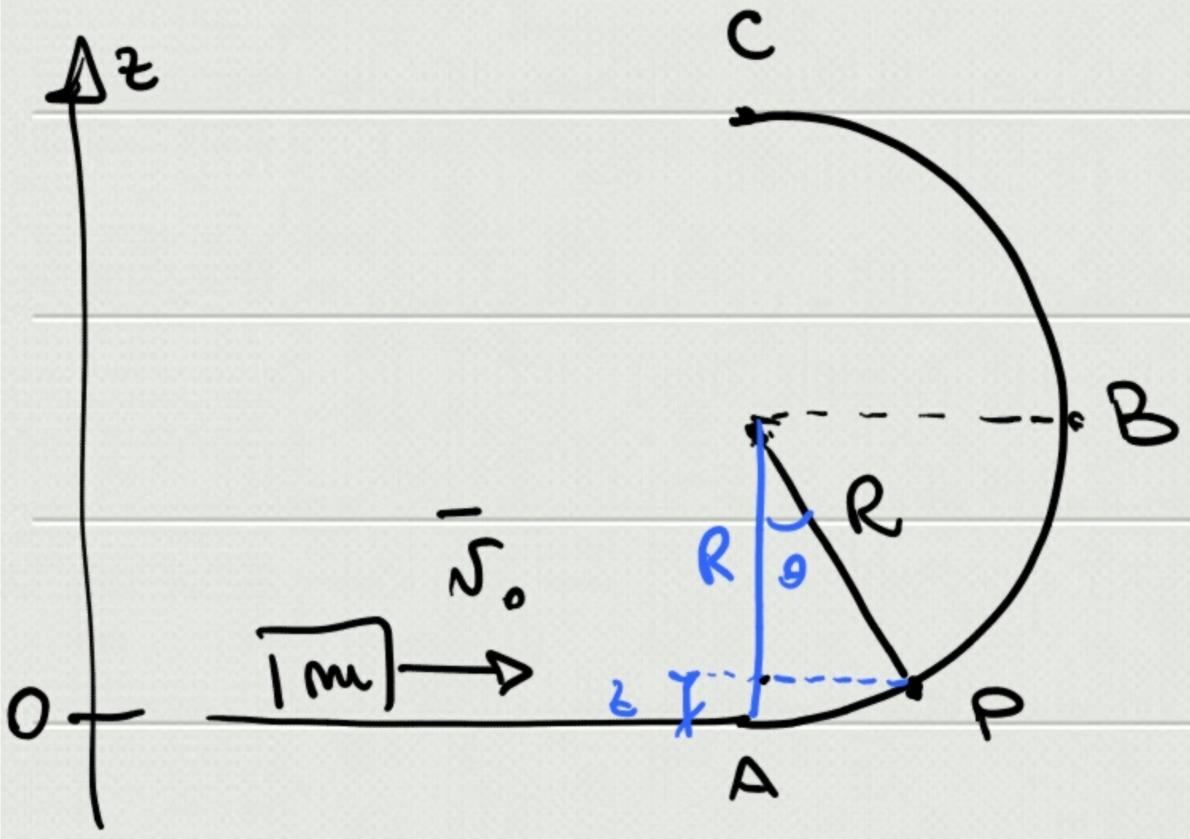
$\hookrightarrow h = z_A$

$$\left.\begin{aligned} E_{m,B} &= E_{k,B} + E_{p,B} = \frac{1}{2} m v_B^2 + \emptyset = \frac{1}{2} m v_B^2 \\ &\overset{!}{=} E_{m,A} = mgh \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow v_B = \sqrt{2gh}$$

$$\left.\begin{aligned} E_{m,C} &= E_{k,C} + E_{p,C} = \frac{1}{2} m v_C^2 \\ &\overset{!}{=} E_{m,B} = \frac{1}{2} m v_B^2 \end{aligned}\right\} \Rightarrow v_C = v_B$$

$$\left.\begin{aligned} E_{m,D} &= E_{k,D} + E_{p,D} = \emptyset + mgh' \\ &\overset{!}{=} E_{m,A} = mgh \end{aligned}\right\} \boxed{h' = h}$$



v_B, v_C

N_B, N_C

liscio ⇒ solo forze cons.

$$\boxed{E_m = \text{cost}} \qquad \boxed{E_{m,in} = \frac{1}{2} m v_0^2}$$

$$E_{m,P} = E_{k,P} + E_{pot,P} = \frac{1}{2} m v^2 + mgz$$

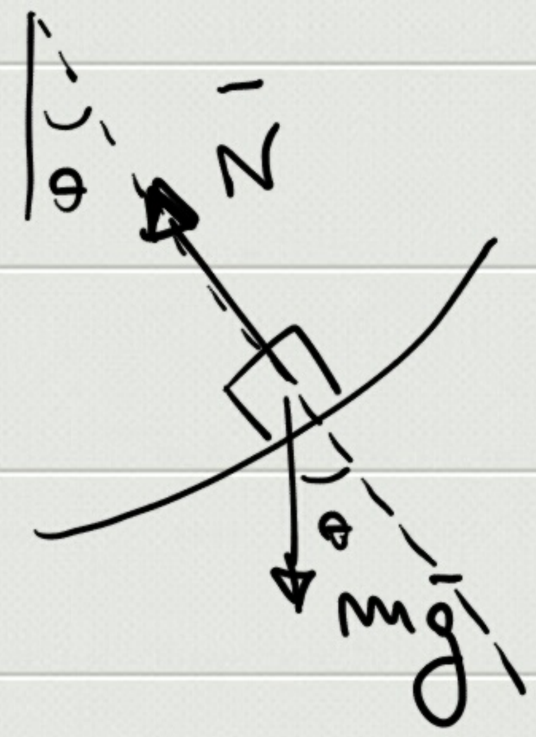
$$z = R - R\cos\theta = R(1-\cos\theta)$$

$$E_{m,P} = \frac{1}{2} m v^2(\theta) + mgR(1-\cos\theta)$$
$$= \frac{1}{2} m v_0^2$$

$$\Rightarrow v(\theta) = \sqrt{v_0^2 - 2gR(1-\cos\theta)}$$

$$v_B = v\left(\theta = \frac{\pi}{2}\right) = \sqrt{v_0^2 - 2gR}$$

$$v_C = v(\theta = \pi) = \sqrt{v_0^2 - 4gR}$$



$$\bar{N} + m\bar{g} = m\bar{a}$$

$$\bar{u}_N : \quad N - mg\cos\theta = m a_N = m\frac{v^2}{R}$$

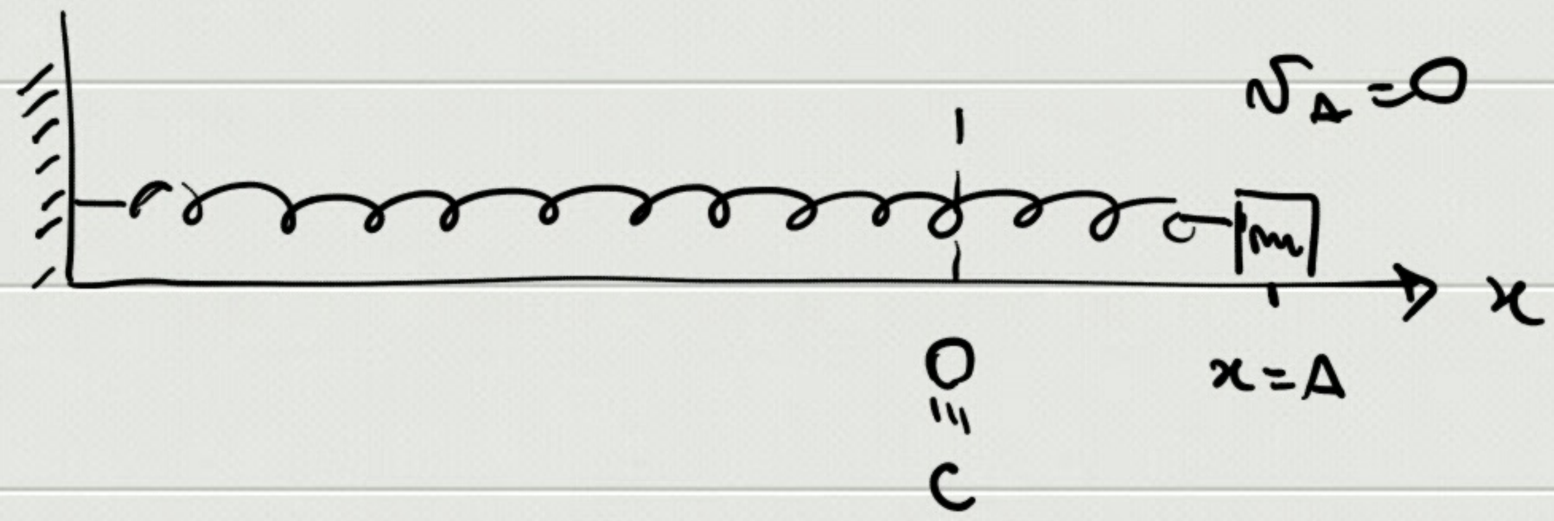
$$\Rightarrow N(\theta) = m\frac{v^2(\theta)}{R} + mg\cos\theta$$

$$N_B = N\left(\theta = \frac{\pi}{2}\right) = \frac{m}{R}\left(v_0^2 - 2gR\right)$$

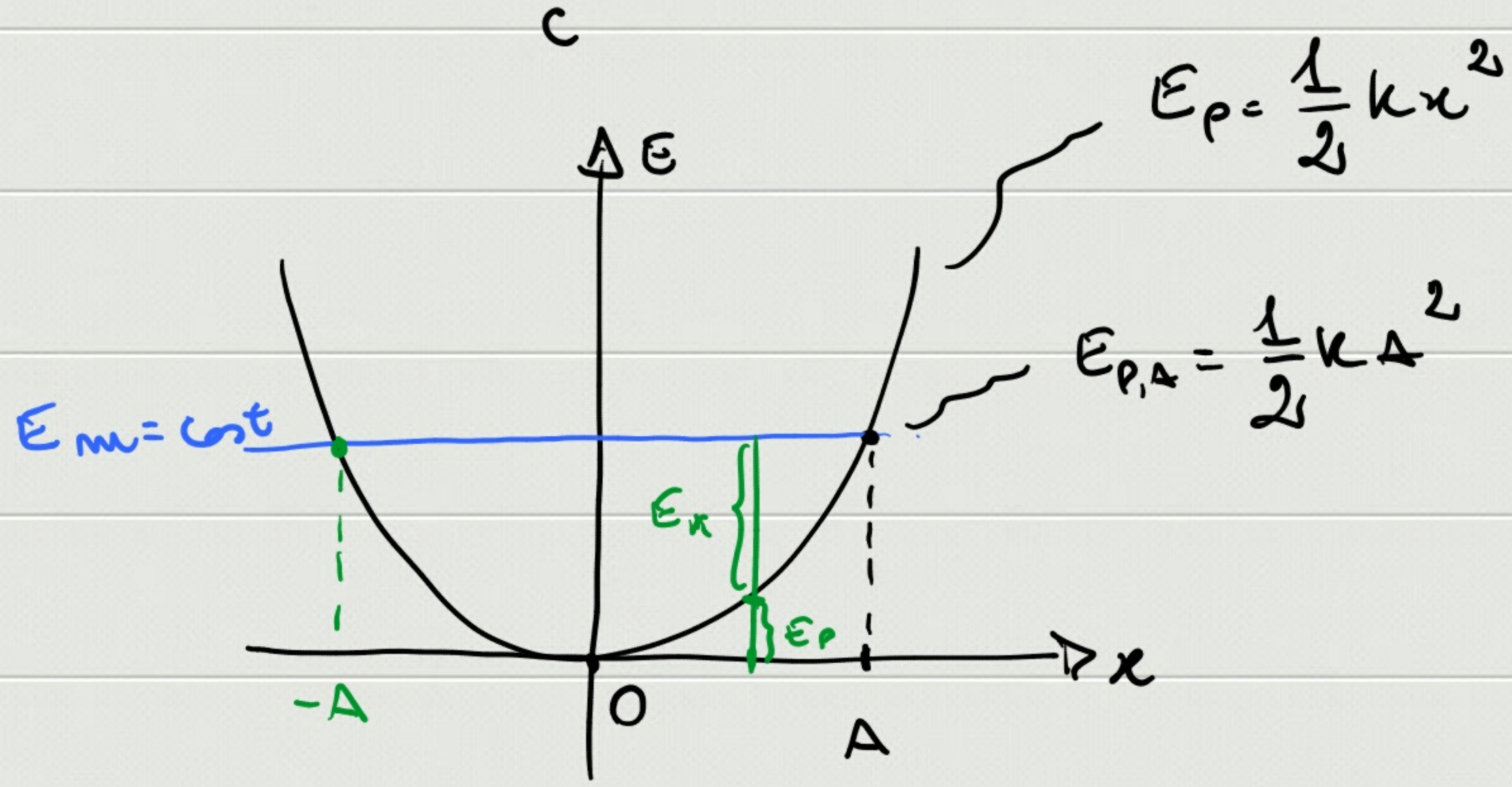
$$N_C = N(\theta = \pi) = \frac{m}{R}\left(v_0^2 - 4gR\right) - mg = m\left(\frac{v_0^2}{R} - 5g\right)$$

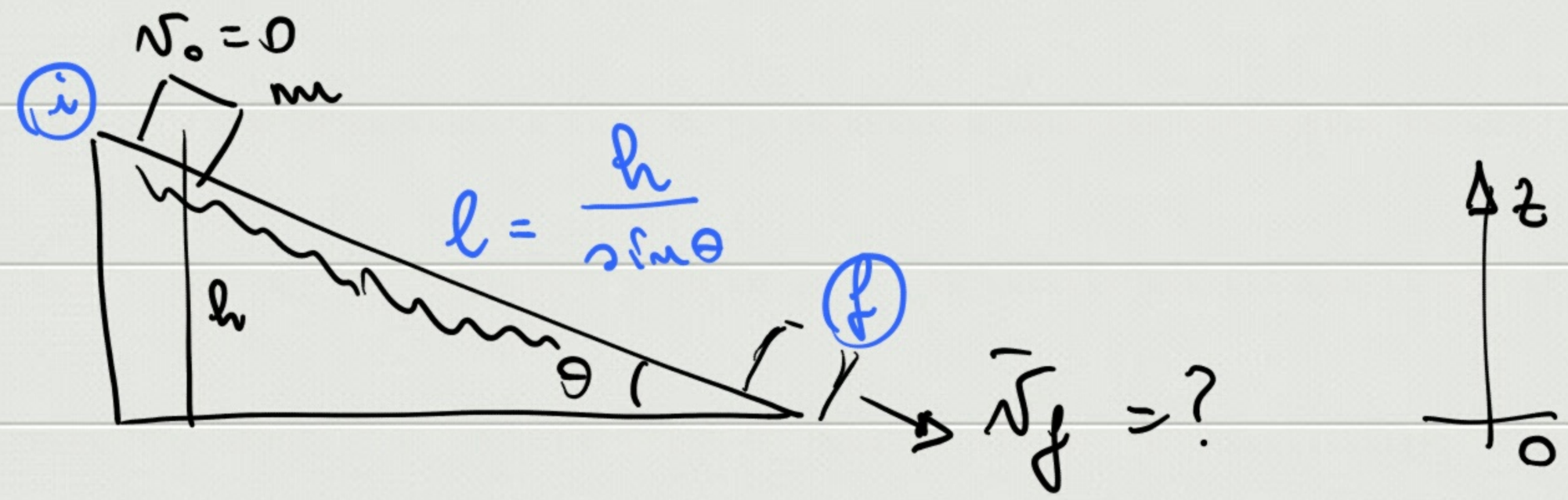
$v_{0,\text{mín}} = ?$ $\quad \Rightarrow N_C = 0$

$$\Rightarrow \underset{(\text{mín})}{v_0} = \sqrt{5gR}$$



linear

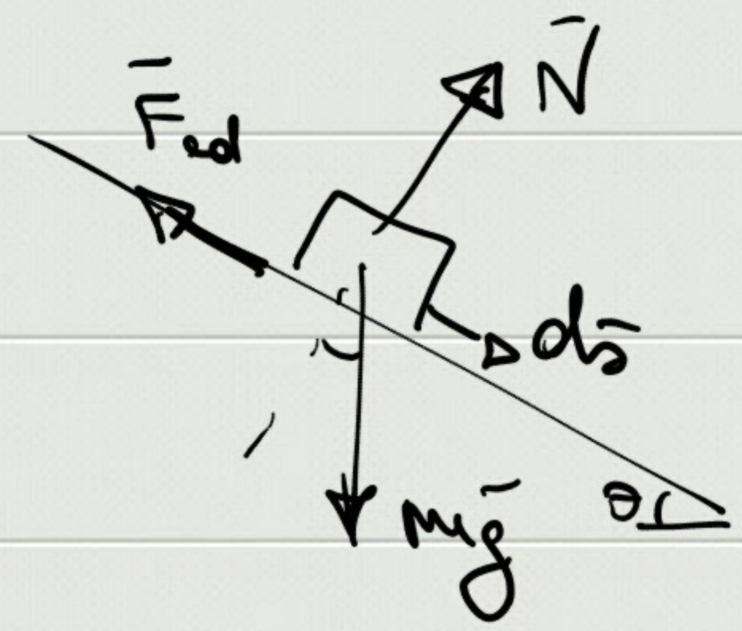




forze non cons. $\Rightarrow$ $\boxed{W_{nc} = \Delta E_m}$

$$E_{m,f} = E_{k,f} + E_{p,f} = \frac{1}{2} m v_f^2$$

$$E_{m,i} = mgh$$



$$W_{m,c} = \int_i^f \bar{F}_{ad}\, d\bar{s} =$$

$$= \int_i^f (-\mu_d N \bar{u}_v)\, d\bar{s} = -\mu_d mg\cos\theta \int_i^f ds =$$

$$= -\mu_d mg\cos\theta \cdot l = -\mu_d mgh \operatorname{cotg}\theta$$

$$-\mu_d \cancel{m} gh \operatorname{ctg}\theta = \frac{1}{2}\cancel{m} v_f^2 - \cancel{m} gh$$

$$\Rightarrow \quad v_f = \sqrt{2gh\,(1 - \mu_d \operatorname{ctg}\theta)}$$